

visStatistics: Automated selection and visualisation of statistical hypothesis tests in R

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Abstract We present **visStatistics**, an R package that automates the selection and visualisation of statistical hypothesis tests between two vectors. Given vectors of class `numeric`, `integer`, or `factor`, the main function `visstat()` determines the appropriate test from the data types, distributional assumptions, and sample size alone, without any manual intervention. Visual outputs — box plots, bar charts, regression lines with confidence bands, mosaic plots, and residual diagnostics — are annotated with test statistics, assumption-check results, and post-hoc comparisons where applicable. The decision logic follows the general linear model framework, assessing normality on pooled model residuals rather than per group, which preserves statistical power. The package is particularly suited for server-side R applications where end users interact exclusively via a web interface, and for time-constrained settings such as statistical consulting or introductory teaching. All tests are applied using their default settings from base R, keeping the dependency footprint minimal and the decision tree transparent.

1 Introduction

Most routine data analyses reduce to a comparatively small set of inferential frameworks: group comparisons, contingency-table analyses, correlation, and simple regression (Sato et al., 2017; Chicco et al., 2025). Selecting the appropriate test within each framework, however, requires evaluating variable types, sample sizes, and distributional assumptions — a process that is error-prone, time-consuming, and a recurring bottleneck in statistical consulting and undergraduate teaching.

Several R packages address parts of this problem. **compareGroups** (Subirana et al., 2014) automates test choice via a per-group Shapiro–Wilk normality test and produces publication-ready descriptive tables, but its primary output is tabular rather than graphical, and assumption diagnostics are not displayed alongside results. **ggstatsplot** (Patil, 2021) produces information-rich **ggplot2**-based (Wickham, 2016) plots with embedded test results, but requires the user to select the appropriate plotting function for each data-type combination (e.g., `ggbetweenstats()` for group comparisons, `ggscatterstats()` for regression). **rstatix** (Kassambara, 2025) provides a pipe-friendly framework for common tests with tidy output, and **ggpubr** (Kassambara, 2026) adds statistical annotations to publication-ready plots; both leave test selection entirely to the user.

visStatistics takes a different approach: a single function `visstat(x, y)` accepts any combination of numeric and categorical vectors and resolves the full decision pipeline — test selection, execution, assumption checking, and visualisation — without any further input. Three design choices distinguish it from the packages above.

First, normality is assessed on the pooled residuals of a linear model $\text{lm}(y \sim x)$, following the general linear model (GLM) framework (Searle, 1971), rather than per group. Group-wise testing is statistically less powerful and more likely to reject normality unnecessarily, pushing the analysis to non-parametric methods (Lumley et al., 2002). When heteroscedasticity is subsequently detected, group-wise normality diagnostics are shown additionally, since Welch’s methods use group-specific variance estimates.

Second, assumption diagnostics are always displayed, regardless of whether assumptions are met. This enables users to visually override the automated decision in individual cases, and supports the pedagogical goal of making assumption checking an integral part of the analysis rather than an afterthought.

Third, all tests use their default settings from base R functions, keeping the dependency footprint minimal and the decision tree transparent. The package does not wrap external statistical engines; the automated layer is purely in the routing logic.

2 Package overview

visStatistics is available on CRAN. The package accepts input in three equivalent forms:

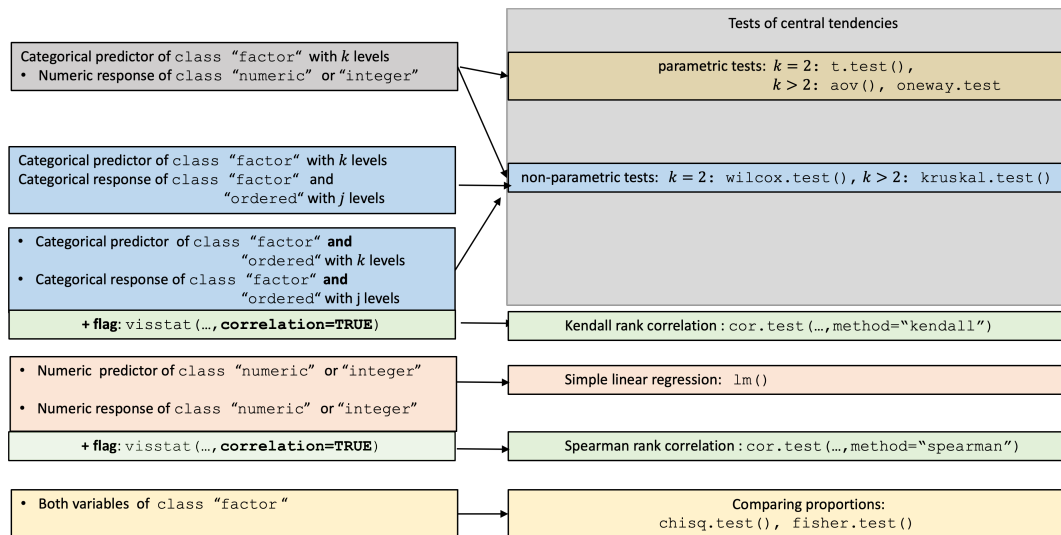


Figure 1: Overview of all implemented tests by input class.

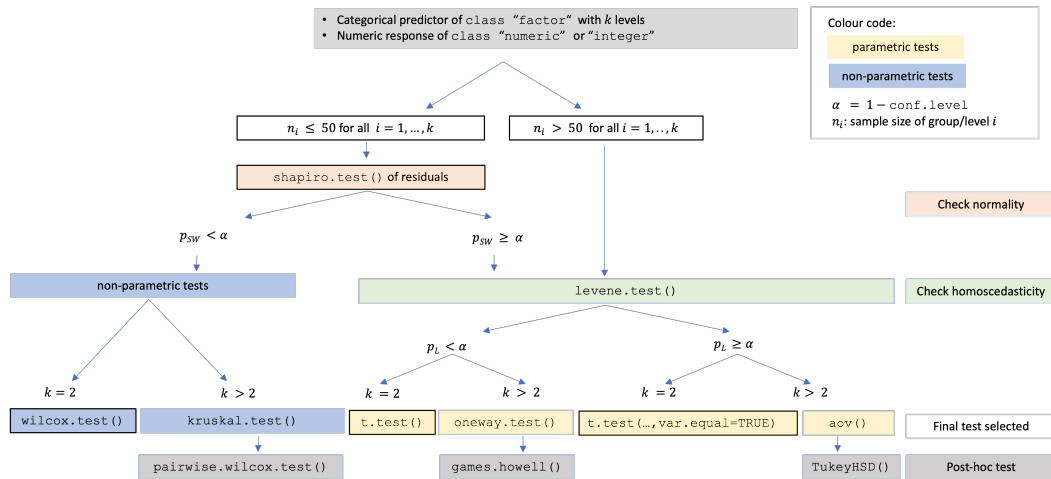


Figure 2: Decision tree for a numeric response and categorical predictor, based on sample size, normality of residuals, and homoscedasticity.

Recommended form:

```
visstat(x, y)
```

Formula interface:

```
visstat(y ~ x, data = dataframe)
```

Backward-compatible form:

```
visstat(dataframe, "namey", "namex")
```

where x and y are vectors of class "numeric", "integer", or "factor". The function returns an object of class "visstat" with `print()`, `summary()`, and `plot()` S3 methods, and optionally writes graphics to disk via [Cairo](#) (Urbanek and Horner, 2025).

3 Decision logic

The significance level $\alpha = 1 - \text{conf.level}$ is user-controlled (default 0.95). Figure 1 gives a high-level overview of all implemented tests by input class; Figure 2 details the decision path for a numeric response and categorical predictor. Technical definitions of all test statistics are collected in [Appendix B](#).

3.1 Numeric response and categorical predictor: comparing central tendencies

A linear model $\text{lm}(y \sim x)$ is fitted and assumption diagnostics are always displayed via `vis_lm_assumptions()` (see [Appendix A](#)). The Shapiro–Wilk test ([Shapiro and Wilk, 1965](#)) is applied to internally studentised residuals from `rstandard()` ([Cook and Weisberg, 1982](#)). When all groups contain more than 50 observations, normality is assumed by the central limit theorem ([Lumley et al., 2002](#); [Rasch et al., 2011](#)).

If normality is rejected ($p_{\text{SW}} \leq \alpha$), non-parametric tests are selected: `wilcox.test()` ([Mann and Whitney, 1947](#)) for two groups, or `kruskal.test()` ([Kruskal and Wallis, 1952](#)) followed by Holm-adjusted pairwise `wilcox.test()` ([Holm, 1979](#)) for more than two groups.

If normality is not rejected, the Levene–Brown–Forsythe test `levene.test()` ([Brown and Forsythe, 1974](#)) (see [Appendix B](#)) assesses variance homogeneity. For homoscedastic data, Student’s `t.test(var.equal = TRUE)` or Fisher’s `aov()` ([Fisher and Yates, 1990](#)) with `TukeyHSD()` ([Hochberg and Tamhane, 1987](#)) is applied. For heteroscedastic data, Welch’s `t.test()` ([Welch, 1947](#); [Satterthwaite, 1946](#)) or `oneway.test()` ([Welch, 1951](#)) with `games.howell()` ([Games and Howell, 1976](#)) is applied. When switching to Welch methods, group-wise normality diagnostics are additionally displayed via `vis_group_normality()`, since the pooled `lm()` residuals are no longer the appropriate diagnostic under heteroscedasticity.

3.2 Both variables numeric: regression or rank correlation

By default, `visstat()` fits a simple linear regression via `lm()`, regardless of whether GLM assumptions are met, and always displays assumption diagnostics. When `correlation = TRUE` is set explicitly, Spearman’s ρ is computed via `cor.test(..., method = "spearman")`. Because the choice between modelling a directional relationship and measuring a monotone association cannot be automated from data types alone, Spearman correlation is never triggered automatically.

3.3 Both variables categorical: comparing proportions

When both variables are categorical, `visstat()` tests independence using Pearson’s χ^2 test (`chisq.test()`) or Fisher’s exact test (`fisher.test()`), depending on expected cell counts following Cochran’s rule ([Cochran, 1954](#)): the χ^2 approximation is used if no expected cell count is less than 1 and no more than 20% of cells have expected counts below 5. Yates’ continuity correction ([Yates, 1934](#)) is applied by default to 2×2 tables. For general $R \times C$ tables, Pearson’s χ^2 result is supplemented with a mosaic plot from `vcd` ([Meyer et al., 2006, 2024](#)) with tiles coloured by Pearson residuals.

3.4 Ordered factors

When the response is an ordered factor, `visstat()` converts it to integer level codes and follows the non-parametric path. When both variables are ordered and `correlation = TRUE`, Kendall’s τ_b ([Kendall, 1945](#); [Agresti, 2010](#)) is computed instead, as it corrects for ties explicitly — unavoidable with few ordinal levels ([Xu et al., 2013](#)).

4 Examples

The number of plots returned by `visstat()` depends on the input types and the outcome of assumption checks. Tests that have no separate assumption diagnostics — Kendall’s τ_b for two ordered factors — produce a single plot. Most tests produce two panels: an assumption-diagnostic panel and a result panel. When heteroscedasticity is detected and `visstat()` routes to a Welch method, a third panel with group-wise normality diagnostics is added, since pooled-residual diagnostics are no longer appropriate under unequal variances.

4.1 Two groups: Welch’s t-test

With binary factor `am` (transmission type, 0 = automatic, 1 = manual) and continuous response `mpg` (fuel efficiency) from the `mtcars` dataset, `visstat()` fits `lm(mpg ~ am)`, applies Shapiro–Wilk to the pooled residuals (not rejected), and subsequently the Levene–Brown–Forsythe test, which detects heteroscedasticity. The routing therefore leads to Welch’s t-test rather than Student’s t-test. Because the pooled-residual diagnostic is no longer appropriate under heteroscedasticity, `vis_group_normality()` additionally displays per-group normality checks, yielding three panels in total.

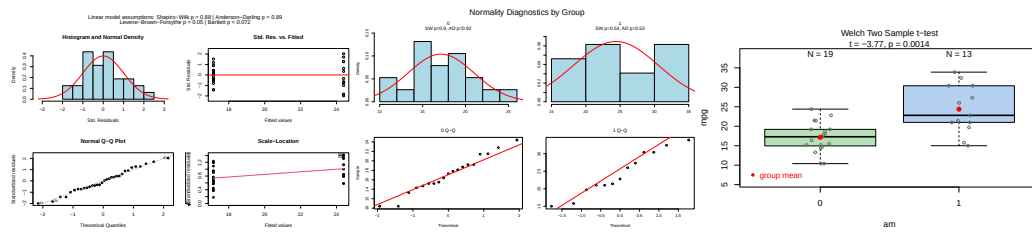


Figure 3: Welch's t-test applied to the mtcars dataset (am vs. mpg). Pooled-residual assumption diagnostics (Shapiro–Wilk, Bartlett, Levene), group-wise normality diagnostics, and box plots of fuel efficiency by transmission type with the Welch t-test result.

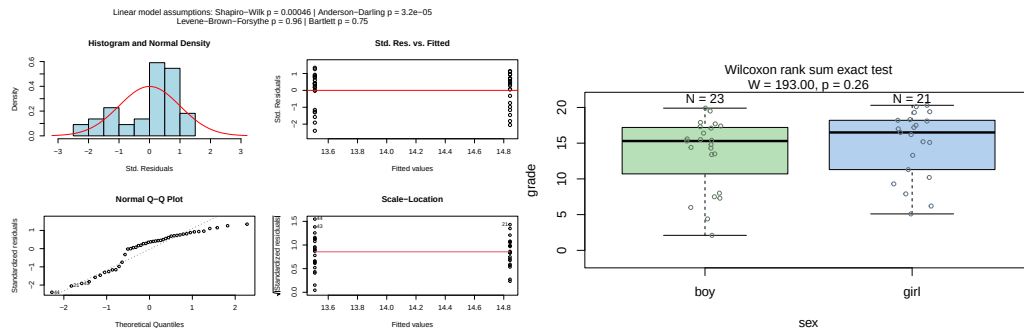


Figure 4: Wilcoxon rank-sum test applied to examination grades by sex. Assumption diagnostics (Shapiro–Wilk rejected; non-parametric path selected) and box plots with the Wilcoxon test result.

```
mtcars$am <- as.factor(mtcars$am)
t_test_stats <- visstat(mtcars$am, mtcars$mpg)
```

4.2 Two groups: Wilcoxon rank-sum test

The grades_gender dataset records examination grades for a class of 44 students (21 girls, 23 boys). After fitting `lm(grade ~ sex)`, Shapiro–Wilk rejects normality of the pooled residuals, routing the analysis to the non-parametric Wilcoxon rank-sum test.

```
grades_gender <- data.frame(
  sex = as.factor(c(rep("girl", 21), rep("boy", 23))),
  grade = c(19.3, 18.1, 15.2, 18.3, 7.9, 6.2, 19.4, 20.3, 9.3, 11.3,
            18.2, 17.5, 10.2, 20.1, 13.3, 17.2, 15.1, 16.2, 17.0, 16.5,
            5.1, 15.3, 17.1, 14.8, 15.4, 14.4, 7.5, 15.5, 6.0, 17.4,
            7.3, 14.3, 13.5, 8.0, 19.5, 13.4, 17.9, 17.7, 16.4, 15.6,
            17.3, 19.9, 4.4, 2.1)
)
wilcoxon_stats <- visstat(grades_gender$sex, grades_gender$grade)
```

4.3 Multiple groups: one-way ANOVA with Tukey HSD

With factor block (experimental block) and continuous response yield from the npk pea-growing dataset, Shapiro–Wilk does not reject normality of pooled residuals and the Levene test does not reject homoscedasticity. Fisher's one-way ANOVA is therefore applied, followed by Tukey HSD post-hoc comparisons with a compact letter display.

```
anova_npk <- visstat(npk$block, npk$yield, conf.level = 0.95)
```

4.4 Multiple groups: Kruskal-Wallis rank sum test

Applied to petal widths of three iris species from the built-in iris dataset, Shapiro–Wilk strongly rejects normality of the pooled residuals, routing the analysis to `kruskal.test()` followed by Holm-adjusted pairwise Wilcoxon post-hoc comparisons. All three pairwise comparisons are significant.

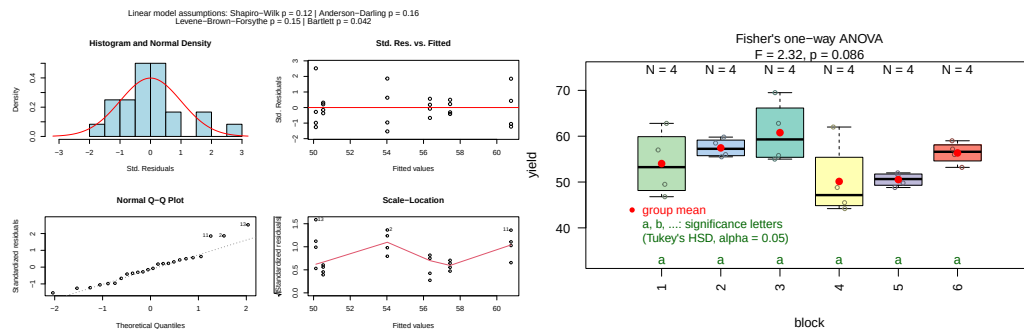


Figure 5: Fisher's one-way ANOVA applied to the npk dataset (block vs. yield). Assumption diagnostics and box plots of pea yield by experimental block; compact letter display (TukeyHSD, $\alpha = 0.05$) shows no significant pairwise differences.

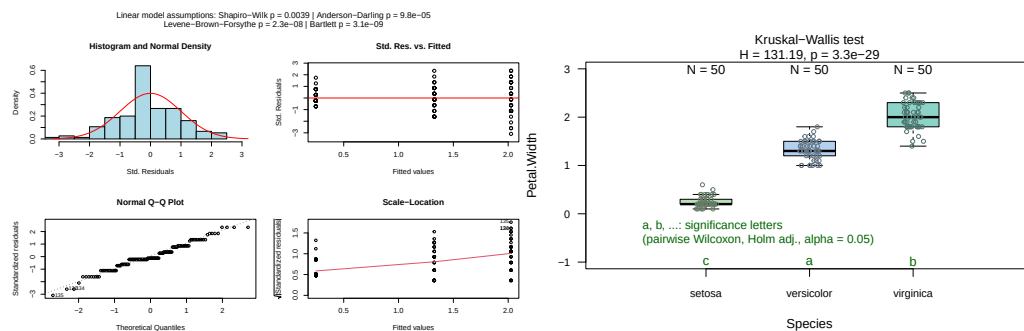


Figure 6: Kruskal-Wallis test applied to the iris dataset (Species vs. Petal.Width). Assumption diagnostics (Shapiro-Wilk rejected) and box plots with the Kruskal-Wallis result; all pairwise post-hoc comparisons significant (all letters differ).

```
kruskal_iris <- visstat(iris$Species, iris$Petal.Width)
```

4.5 Numeric response and numeric predictor: simple linear regression

The swiss dataset records standardised fertility and socioeconomic indicators for 47 French-speaking Swiss provinces in 1888. We examine how the share of draftees achieving the highest army examination score (Examination) predicts the fertility measure (Fertility), with `conf.level = 0.99`. Both normality tests pass and the Breusch-Pagan test confirms homoscedasticity, validating the linear model.

```
linreg_swiss <- visstat(swiss$Examination, swiss$Fertility, conf.level = 0.99)
```

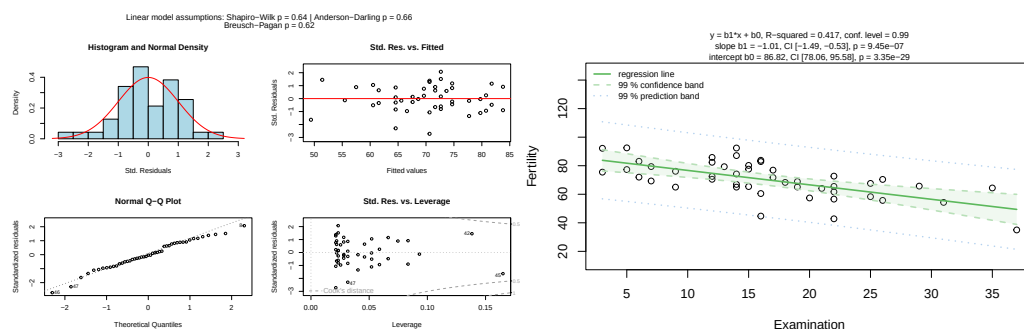


Figure 7: Simple linear regression of Fertility on Examination for the swiss dataset (`conf.level = 0.99`). Assumption diagnostics (Shapiro-Wilk, Anderson-Darling, Breusch-Pagan) and scatter plot with fitted regression line, 99% confidence band (dark shading), and 99% prediction band (light shading).

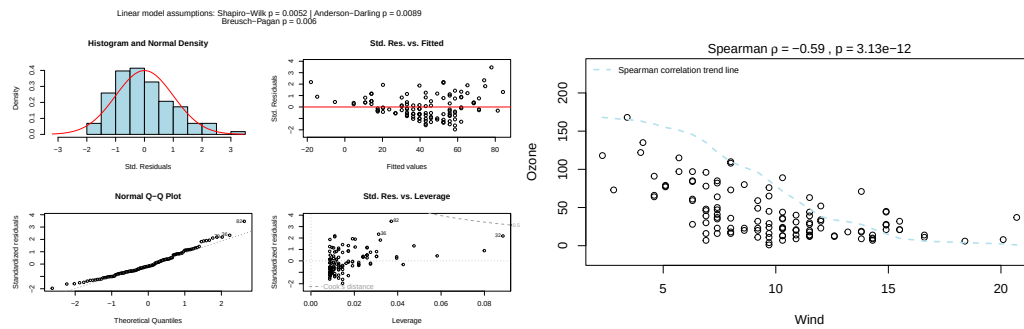


Figure 8: Spearman rank correlation of Wind and Ozone from the airquality dataset (correlation = TRUE). Assumption diagnostics and scatter plot of ranked values annotated with ρ and the p -value.

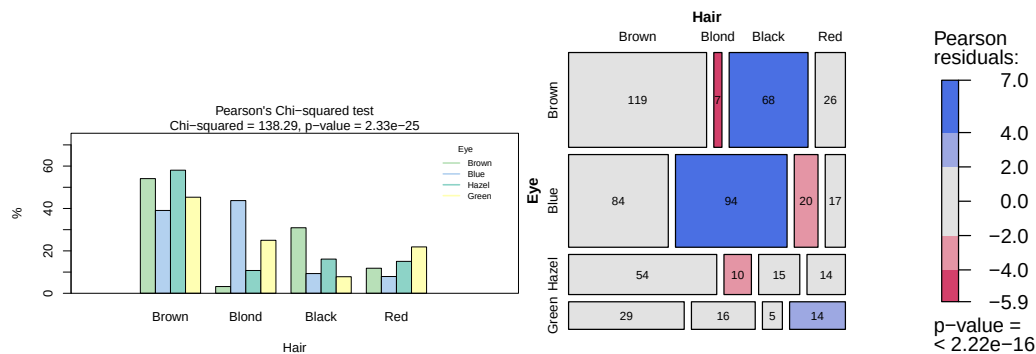


Figure 9: Pearson's χ^2 test applied to the HairEyeColor dataset. Grouped bar chart of eye colour by hair colour and mosaic plot with tiles coloured by Pearson residuals (blue: over-represented, red: under-represented).

4.6 Two numeric variables: Spearman rank correlation

With wind speed (Wind) and ozone concentration (Ozone) from the airquality dataset (Chambers, 2018), a default `visstat()` call fits linear regression but flags violated normality and homoscedasticity. Setting `correlation = TRUE` switches to Spearman's ρ , which measures the monotone association on ranks without distributional assumptions.

```
spearman_air <- visstat(airquality$Wind, airquality$Ozone, correlation = TRUE)
```

4.7 Two categorical variables: Pearson's χ^2 test

With Eye and Hair from the HairEyeColor dataset, all expected cell counts exceed the Cochran thresholds (Cochran, 1954), so the χ^2 approximation is used. For this 4×4 table, `visstat()` returns both a bar chart of row percentages and a mosaic plot with tiles coloured by Pearson residuals, making the cells driving the association immediately visible.

```
hair_eye_df <- counts_to_cases(as.data.frame(HairEyeColor))
visstat(hair_eye_df$Eye, hair_eye_df$Hair)
```

4.8 Two categorical variables: Fisher's exact test

Restricting HairEyeColor to male participants with black or brown hair and hazel or green eyes yields a 2×2 table where one cell's expected count falls below 5, violating Cochran's rule (Cochran, 1954). `visstat()` therefore applies Fisher's exact test. For 2×2 tables, the returned object also contains the conditional maximum likelihood estimate of the odds ratio and its confidence interval.

```
hair_eye_male <- HairEyeColor[, , 1]
black_brown_hazel_green <- hair_eye_male[1:2, 3:4]
black_brown_hazel_green_df <- counts_to_cases(
```

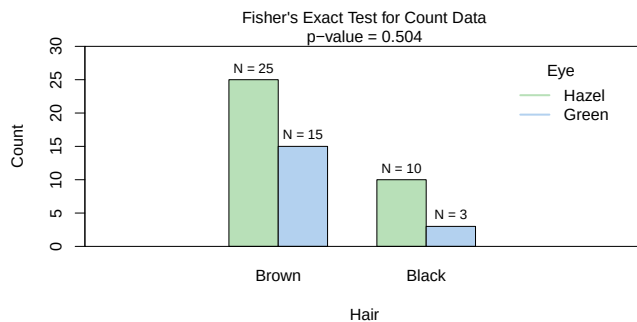


Figure 10: Fisher's exact test applied to a 2×2 subset of HairEyeColor (male participants, black/brown hair, hazel/green eyes). Bar chart of absolute counts with the p -value; the odds ratio and confidence interval are accessible from the returned object.

```
as.data.frame(black_brown_hazel_green))
fisher_stats <- visstat(black_brown_hazel_green_df$Eye,
                       black_brown_hazel_green_df$Hair)
fisher_stats$estimate # odds ratio

#> odds ratio
#> 0.5062015

fisher_stats$conf.int # 95 % CI

#> [1] 0.07725895 2.40885255
#> attr("conf.level")
#> [1] 0.95
```

4.9 Two ordered variables: Kendall's τ_b

When both variables are ordered factors and `correlation = TRUE` is set, `visstat()` tests for a monotone association via Kendall's τ_b (Kendall, 1945). A hypothetical survey of 150 secondary-school students records alcohol consumption frequency and academic performance on five-point ordinal scales; a negative association is induced by construction.

```
set.seed(42)
n <- 150
xs <- sample(1:5, n, replace = TRUE)
ys <- pmin(5, pmax(1, (6 - xs) + sample(-1:1, n, replace = TRUE)))
likert_alc <- c("never", "rarely", "sometimes", "often", "always")
likert_perf <- c("poor", "fair", "ok", "good", "great")
alcohol <- ordered(likert_alc[xs], levels = likert_alc)
performance <- ordered(likert_perf[ys], levels = likert_perf)
kendall_result <- visstat(performance, alcohol, correlation = TRUE)
```

5 Saving graphical output

All plots can be saved in any format supported by [Cairo](#) (Urbanek and Horner, 2025) — "png", "pdf", "svg", "ps", "tiff" — by supplying the `graphicsoutput` and `plotDirectory` arguments. File names follow the pattern `testname_namey_namex.ext` by default, or a user-supplied `plotName`. The full file paths of generated graphics are stored as the `"plot_paths"` attribute on the returned `"visstat"` object.

6 Limitations

6.1 Scope of automated test selection

All tests are applied with their default settings from base R. As a consequence, paired tests are not supported and interaction terms are not available in the ANOVA path. Only simple linear

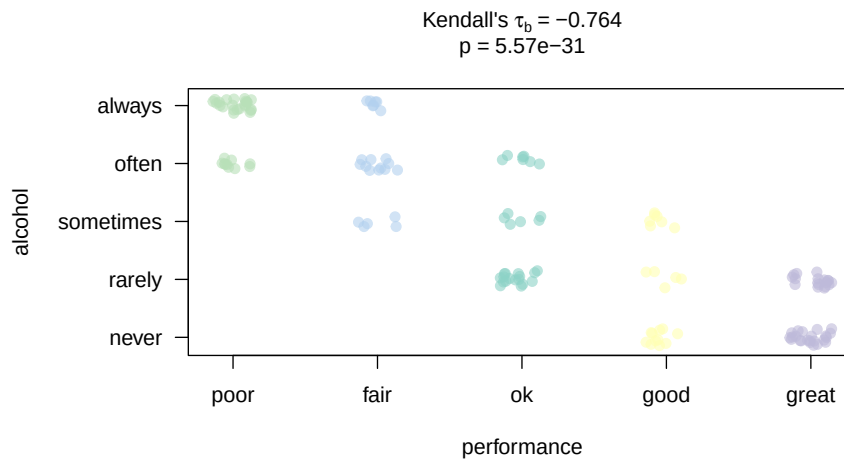


Figure 11: Kendall's τ_b for a hypothetical survey ($n = 150$): alcohol consumption frequency vs. academic performance (both five-point ordinal scales). Jittered rank-rank scatter annotated with τ_b and the p -value; the negative monotone association is significant.

regression is implemented, as multiple regression cannot be visualised in two dimensions. When regression assumptions are violated, [visStatistics](#) offers Spearman rank correlation (`correlation = TRUE`) as the sole automated alternative. Alternative methods such as data transformations, generalised linear models, or robust regression are not implemented: each requires user judgment — about the transformation family, the link function, or the estimator — that cannot be automated without substantially expanding the decision tree and increasing the risk of Type I error inflation.

6.2 P-value-based assumption testing

No single test maintains optimal Type I error rates and power across all distributions ([Olejnik and Algina, 1987](#)). Normality tests behave poorly at both extremes of the sample-size range: they fail to detect non-normality in small samples and flag negligible departures as significant in large ones ([Ghasemi and Zahediasl, 2012](#); [Fagerland, 2012](#)). The $n > 50$ per-group threshold for bypassing the Shapiro–Wilk test is necessarily arbitrary; simulation studies indicate that the required sample size for the central limit theorem to take effect depends on the skewness and kurtosis of the underlying distribution ([Fagerland, 2012](#)). Assumption tests provide no information on the nature of the deviation ([Shatz, 2024](#)), and combining multiple tests inflates the overall Type I error rate. For these reasons, the diagnostic plots generated by `visstat()` should always be inspected alongside the automated decision; the `conf.level` argument provides a direct handle on the α used throughout.

6.3 Visualisation

The package depends exclusively on base R graphics with no [ggplot2](#) ([Wickham, 2016](#)) dependency, keeping the transitive dependency footprint minimal. For more polished, annotated plots of chosen statistical tests, see [ggstatsplot](#) ([Patil, 2021](#)) or [ggpubr](#) ([Kassambara, 2026](#)).

7 Summary

[visStatistics](#) provides a single-function interface to automated statistical test selection, execution, and visualisation for the most commonly used inferential frameworks in two-variable analyses. Its principal contribution is the integration of the full pipeline — data-type detection, distributional assumption checking on GLM residuals, test routing, and annotated graphical output — into a reproducible, server-deployable workflow with a minimal dependency footprint. The package is available on CRAN at <https://CRAN.R-project.org/package=visStatistics> and on GitHub at <https://github.com/shhschilling/visStatistics>.

8 Appendix A: The general linear model

The general linear model (Searle, 1971) provides a unified framework underlying Student's t-test, Fisher's ANOVA, and simple linear regression:

$$Y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_{k-1} x_{i,k-1} + \varepsilon_i, \quad \varepsilon_i \sim N(0, \sigma^2).$$

Student's t-test uses one binary indicator variable x_{i1} ; testing $H_0 : \beta_1 = 0$ is equivalent to testing $H_0 : \mu_1 = \mu_2$.

Fisher's ANOVA uses $k - 1$ indicator variables for k groups; testing $H_0 : \beta_1 = \cdots = \beta_{k-1} = 0$ is equivalent to testing equality of all group means.

Simple linear regression uses one continuous predictor; testing $H_0 : \beta_1 = 0$ examines whether a linear relationship exists.

The two-sample case connects t-tests and ANOVA: $t^2 = F$. For two groups, `t.test(var.equal = TRUE)` and `aov()` yield identical p -values, as do `t.test(var.equal = FALSE)` and `oneway.test()`.

9 Appendix B: Test statistics

9.1 Normality tests

Both normality tests are applied to the internally studentised residuals $r_i^* = e_i / (s\sqrt{1 - h_{ii}})$, where e_i are the raw residuals, s^2 is the residual mean square, and h_{ii} is the leverage of observation i (Cook and Weisberg, 1982).

Shapiro–Wilk test `shapiro.test()`

The Shapiro–Wilk test (Shapiro and Wilk, 1965) evaluates whether a sample $x_1, \dots, x_n$ comes from a normal distribution. Let $x_{(i)}$ denote the i -th order statistic. The test statistic is

$$W = \frac{\left(\sum_{i=1}^n a_i x_{(i)} \right)^2}{\sum_{i=1}^n (x_i - \bar{x})^2},$$

where the constants a_i are derived from the means, variances, and covariances of the order statistics of a standard normal sample (Shapiro and Wilk, 1965). W takes values in $(0, 1]$; values close to 1 indicate normality. Simulation studies show that the Shapiro–Wilk test has the highest power among normality tests for small to moderate sample sizes ($n = 10$ to 100) (Razali and Wah, 2011). It drives the parametric/non-parametric branching decision in `visstat()`; for groups with more than 50 observations normality is assumed by the central limit theorem (Lumley et al., 2002; Rasch et al., 2011).

Anderson–Darling test `ad.test()`

The Anderson–Darling test (Anderson and Darling, 1952) is particularly sensitive to deviations in the tails of the distribution (Razali and Wah, 2011; Yap and Sim, 2011). Let $z_i = (x_{(i)} - \bar{x})/s$ be the standardised order statistics and Φ the standard normal CDF. The test statistic is

$$A^2 = -n - \frac{1}{n} \sum_{i=1}^n (2i - 1) [\ln \Phi(z_i) + \ln(1 - \Phi(z_{n+1-i}))].$$

The implementation uses `ad.test()` from `nortest` (Gross and Ligges, 2015). The Anderson–Darling result is shown in all assumption plots but does **not** drive the branching decision; only Shapiro–Wilk does.

9.2 Homoscedasticity tests

The Levene–Brown–Forsythe test `levene.test()`

The test (Brown and Forsythe, 1974) improves upon Levene’s original proposal (Levene, 1960) by centring on the group median rather than the group mean, making it robust to skewed data and outliers (Allingham and Rayner, 2012). The implementation `levene.test()` mimics the default behaviour of `leveneTest()` in `car` (Fox and Weisberg, 2019) and is the sole driver of the parametric/Welch branching decision.

For each observation y_{ij} in group i , the absolute deviation from the group median $\tilde{y}_i$ is

$$z_{ij} = |y_{ij} - \tilde{y}_i|.$$

The test statistic is the F -statistic from a one-way ANOVA on the z_{ij} values:

$$F = \frac{(N - k) \sum_{i=1}^k n_i (\bar{z}_i - \bar{z})^2}{(k - 1) \sum_{i=1}^k \sum_{j=1}^{n_i} (z_{ij} - \bar{z}_i)^2},$$

where k is the number of groups, $N = \sum_{i=1}^k n_i$ is the total sample size, n_i is the sample size of group i , $\bar{z}_i$ is the mean of the absolute deviations within group i , and $\bar{z}$ is the overall mean of all absolute deviations. Under the null hypothesis of equal variances the statistic follows $F(k - 1, N - k)$.

Bartlett’s test `bartlett.test()`

Bartlett’s test (Bartlett, 1937) has greater power than the Brown–Forsythe test when normality holds (Allingham and Rayner, 2012). Its test statistic is

$$K^2 = \frac{(N - k) \ln s_p^2 - \sum_{i=1}^k (n_i - 1) \ln s_i^2}{1 + \frac{1}{3(k - 1)} \left(\sum_{i=1}^k \frac{1}{n_i - 1} - \frac{1}{N - k} \right)},$$

where s_i^2 is the sample variance of group i and s_p^2 is the pooled variance

$$s_p^2 = \frac{1}{N - k} \sum_{i=1}^k (n_i - 1) s_i^2.$$

Under the null hypothesis the statistic approximately follows $\chi^2(k - 1)$. Bartlett’s result is shown in the assumption plot but does **not** drive the branching decision; only `levene.test()` does.

Breusch–Pagan test `bp.test()`

For simple linear regression, group-based variance tests are not applicable. The Breusch–Pagan test (Breusch and Pagan, 1979) assesses whether the variance of the residuals from the regression model depends on the predictor values. Its result is displayed in the regression assumption plot.

9.3 Student’s t-test and Welch’s t-test

The test statistic for Student’s t-test (`t.test(var.equal = TRUE)`) is

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}},$$

where $\bar{x}_1$ and $\bar{x}_2$ are the sample means, n_1 and n_2 the sample sizes, and s_p the pooled standard deviation

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}},$$

with s_1^2 and s_2^2 the sample variances. The statistic follows a t -distribution with $\nu = n_1 + n_2 - 2$ degrees of freedom.

Welch's t -test relaxes the homoscedasticity assumption. Its statistic is

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_1^2/n_1 + s_2^2/n_2}},$$

with degrees of freedom approximated by the Welch–Satterthwaite equation (Welch, 1947; Satterthwaite, 1946):

$$\nu \approx \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}}.$$

Welch's methods outperform their classical counterparts when variances differ (Moser and Stevens, 1992; Fagerland and Sandvik, 2009; Delacre et al., 2017).

9.4 Wilcoxon rank-sum test

The two-level factor x defines two groups with sample sizes n_1 and n_2 . All $N = n_1 + n_2$ observations are pooled and assigned ranks 1 to N . Let $W_1 = \sum_{i=1}^{n_1} R(x_{1,i})$ denote the rank sum of the first group, where $R(x_{1,i})$ is the rank of observation $x_{1,i}$ in the pooled sample. The test statistic returned by `wilcox.test()` (Mann and Whitney, 1947) is the Mann–Whitney U statistic of the first group:

$$W = U_1 = W_1 - \frac{n_1(n_1 + 1)}{2}.$$

An exact p -value is computed when both groups contain fewer than 50 observations and the data contain no ties; otherwise a normal approximation with continuity correction is used.

9.5 Fisher's one-way ANOVA and Welch's heteroscedastic ANOVA

Fisher's ANOVA test statistic (Fisher and Yates, 1990) is

$$F = \frac{MS_{\text{between}}}{MS_{\text{within}}} = \frac{SS_{\text{between}}/(k-1)}{SS_{\text{within}}/(N-k)} = \frac{\sum_{i=1}^k n_i(\bar{x}_i - \bar{x})^2 / (k-1)}{\sum_{i=1}^k \sum_{j=1}^{n_i} (x_{ij} - \bar{x}_i)^2 / (N-k)},$$

where k is the number of groups, $N = \sum_{i=1}^k n_i$ is the total sample size, $\bar{x}_i$ is the mean of group i , $\bar{x}$ is the overall mean, and x_{ij} is observation j in group i . Under the null hypothesis $F \sim F(k-1, N-k)$. Post-hoc comparisons use `TukeyHSD()` (Hochberg and Tamhane, 1987), which controls the family-wise error rate via the studentised range distribution (Abdi, 2007).

Welch's heteroscedastic ANOVA (`oneway.test()`) (Welch, 1951) down-weights groups with large variance. Its test statistic is

$$F_W = \frac{\sum_{i=1}^k w_i(\bar{y}_i - \bar{y}_w)^2 / (k-1)}{1 + \frac{2(k-2)}{k^2-1} \sum_{i=1}^k \frac{(1-w_i/w)^2}{n_i-1}},$$

where $w_i = n_i/s_i^2$ are the inverse-variance weights, $w = \sum_{i=1}^k w_i$, and $\bar{y}_w = \sum_{i=1}^k w_i \bar{y}_i / w$ is the weighted grand mean. Degrees of freedom are adjusted via a Satterthwaite-type approximation (Satterthwaite, 1946). Post-hoc comparisons use `games.howell()` (Games and Howell, 1976), which applies separate variance estimates per pair. Compact letter displays are produced by `multcompLetters()`

from [multcompView](#) (Graves et al., 2026).

9.6 Kruskal–Wallis test

When normality is rejected, `kruskal.test()` (Kruskal and Wallis, 1952) tests whether the groups come from the same population based on ranks:

$$H = \frac{12}{N(N+1)} \sum_{i=1}^k n_i (\bar{R}_i - \bar{R})^2,$$

where n_i is the sample size of group i , k is the number of groups, $\bar{R}_i$ is the average rank of group i , N is the total sample size, and $\bar{R} = (N+1)/2$ is the expected average rank under the null hypothesis. The statistic approximately follows $\chi^2(k-1)$. Post-hoc comparisons use `Holm.adjusted.pairwise.wilcox.test()` (Holm, 1979).

9.7 Simple linear regression and Spearman rank correlation

The regression plot displays the fitted line $\hat{y} = b_0 + b_1x$, where b_0 is the intercept and b_1 the slope, with pointwise confidence and prediction bands at the specified `conf.level`. Assumption diagnostics always include the Shapiro–Wilk and Anderson–Darling (Gross and Ligges, 2015) tests for residual normality and the Breusch–Pagan test for homoscedasticity (see Section [Homoscedasticity tests](#)).

When `correlation = TRUE`, Spearman’s ρ is computed as Pearson’s r applied to the ranks:

$$\rho = r(\text{rank}(x), \text{rank}(y)) = \frac{\sum_{i=1}^n (\text{rank}(x_i) - \overline{\text{rank}(x)}) (\text{rank}(y_i) - \overline{\text{rank}(y)})}{\sqrt{\sum_{i=1}^n (\text{rank}(x_i) - \overline{\text{rank}(x)})^2} \sqrt{\sum_{i=1}^n (\text{rank}(y_i) - \overline{\text{rank}(y)})^2}}.$$

No distributional assumptions on the original data are required.

9.8 Pearson’s χ^2 test and Fisher’s exact test

Let O_{ij} and E_{ij} denote the observed and expected frequencies in row i and column j of an $R \times C$ contingency table, where rows index the R levels of the response y and columns the C levels of the predictor x . The Pearson residual for cell (i, j) is

$$r_{ij} = \frac{O_{ij} - E_{ij}}{\sqrt{E_{ij}}}, \quad i = 1, \dots, R, \quad j = 1, \dots, C.$$

The test statistic of Pearson’s χ^2 test (Pearson, 1900) is

$$\chi^2 = \sum_{i=1}^R \sum_{j=1}^C r_{ij}^2 = \sum_{i=1}^R \sum_{j=1}^C \frac{(O_{ij} - E_{ij})^2}{E_{ij}},$$

compared to $\chi^2((R-1)(C-1))$. For 2×2 tables, Yates’ continuity correction (Yates, 1934) is applied by default. For general $R \times C$ tables, `visstat()` supplements the bar chart with a mosaic plot in which tiles are coloured by r_{ij} (blue: positive, red: negative).

Fisher’s exact test (Fisher, 1970) is applied when Cochran’s rule (Cochran, 1954) is violated. It evaluates the exact hypergeometric probability of the observed table given the fixed margins. For 2×2 tables with row totals $n_{i\cdot}$ and column totals $n_{\cdot j}$, this probability is

$$\mathbb{P}(N \mid n_{1\cdot}, n_{2\cdot}, n_{\cdot 1}, n_{\cdot 2}) = \frac{\binom{n_{1\cdot}}{n_{11}} \binom{n_{2\cdot}}{n_{21}}}{\binom{n}{n_{\cdot 1}}},$$

where $n = n_{1\cdot} + n_{2\cdot}$ is the total sample size. For 2×2 tables, `fisher.test()` additionally returns the conditional maximum likelihood estimate of the odds ratio $\widehat{\text{OR}} = n_{11}n_{22}/(n_{12}n_{21})$ and its confidence interval.

9.9 Kendall's τ_b

For two ordinal variables with n joint observations, let C denote the number of concordant pairs and D the number of discordant pairs. Kendall's τ_b (Kendall, 1945) is

$$\tau_b = \frac{C - D}{\sqrt{(n_0 - n_1)(n_0 - n_2)}},$$

where $n_0 = n(n - 1)/2$, $n_1 = \sum_i t_i(t_i - 1)/2$ sums over groups of tied ranks in the response (with t_i the size of the i -th tie group), and n_2 is the analogous quantity for the predictor. The denominator correction ensures $\tau_b \in [-1, 1]$ even in the presence of ties, which Spearman's ρ does not guarantee (Kendall, 1945). τ_b is therefore preferred for ordinal data with few levels, where ties are unavoidable (Agresti, 2010; Xu et al., 2013).

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